Clinical attempt — ClinicaAI

Educational simulation · non-certifying · 30/10/2026, 17:19:47

Case : Shortness of breath after orthopaedic surgery

Status : In progress

Clinical case and vital signs

Ms N., 45, presents to the emergency department with shortness of breath for three days.

- ta : 120/60 mmHg
- fc : 110/min
- fr : 28/min
- spo2 : 96 % on room air
- temperature : 37 °C

Free history taking — complete dialogue

Confirmed : Yes

Student : What brought you to the emergency department today?

Patient : I have been short of breath for three days; that is why I came to the emergency department.

Student : When did your shortness of breath start?

Patient : It started three days ago.

Student : Did it start suddenly or gradually?

Patient : It developed over the last few days, not all at once.

Student : Do you have chest pain when breathing deeply?

Patient : Yes, I sometimes feel a sharp pain on the right side of my chest when I take a deep breath.

Student : Have you had a cough or fever?

Patient : I cough a little, but rarely, and do not bring up much mucus. No, I have not had fever or chills.

Student : Have you noticed swelling or pain in either leg?

Patient : I have not noticed unusual pain in either calf, though my operated knee is still tender, and my operated leg sometimes feels uncomfortable. I have not noticed any new swelling in just one calf.

Student : Have you undergone surgery recently?

Patient : Yes, I had surgery for a tibial plateau fracture one week ago.

Student : Have you been able to walk normally since the operation?

Patient : Since the operation, I have been walking very little and use crutches; I am not in bed all day. I was told to limit weight-bearing on my operated leg.

Student : Are you currently taking any medication?

Patient : I take medications prescribed after the operation, but I do not have the prescription with me.

Student : Have you ever had a pulmonary embolism or deep vein thrombosis?

Patient : No, I have never had deep vein thrombosis or pulmonary embolism before.

2. Investigations

Confirmed : Yes

ECG

Teaching example: no numerical ECG result is supplied for this case. ECG findings may be absent or nonspecific.

D-dimers

Value not documented. Interpretation depends on pretest clinical probability.

Troponin

No documented result

Creatinine and estimated GFR

No documented result

Complete blood count and platelets

No documented result

Thoracic MRI

No documented result

1. Initial assessment

Spontaneous reasoning

The patient is a 45-year-old woman presenting with shortness of breath, tachycardia and tachypnea after recent orthopedic surgery.

My first concern is a possible pulmonary embolism, although pneumonia is another possibility.

I would assess her general condition, check her vital signs and oxygen saturation, and look for signs of respiratory distress.

She appears hemodynamically stable because her blood pressure is normal, but I would continue monitoring her closely.

I would also assess the clinical probability of pulmonary embolism using the Wells score.

Selected answers

- airway
- perfusion
- pretest
- bleed

Justification

—

Section confirmed : Yes

Generative AI feedback : 10 / 20

spontane: 6 · choix: 4 · priorites: 0 · justification: 0

Strengths

- Good spontaneous recognition of the need to check vital signs, oxygen saturation, and assess for respiratory distress.
- Correctly identified tachycardia and tachypnea after recent orthopedic surgery as key features raising suspicion for pulmonary embolism.
- Appropriate selection of the Wells or Geneva score to estimate clinical pretest probability.

Areas to improve

- Ensure priorities and justifications are explicitly written out in the designated text fields rather than relying on spontaneous thoughts.
- Pay closer attention to checking bleeding risks and contraindications if anticoagulation is being considered for suspected pulmonary embolism.

Safety notes

- The student selected 'Check bleeding risks and contraindications', which is a relevant safety check for PE management, though not fully elaborated in the text.
- Failure to provide explicit written priorities and justifications results in a score of zero for those sections, despite good underlying clinical intuition.

Full grading data

3. Diagnosis

Spontaneous hypotheses

My main diagnosis is pulmonary embolism because the patient has shortness of breath, tachycardia, pleuritic chest pain and recent orthopedic surgery with reduced mobility.

I would also consider pneumonia, although the absence of fever makes it less likely.

Another possibility is an acute cardiac condition, but the postoperative context strongly suggests a thromboembolic event.

I would assess the probability of pulmonary embolism and use appropriate imaging to confirm the diagnosis.

Selected hypotheses

- Pulmonary embolism

Primary diagnosis

—

Confirmed : Yes

Generative AI feedback : 7 / 15

spontane: 5 · differentiel: 0 · principal: 0 · arguments: 2

Strengths

- The student spontaneously identified pulmonary embolism and provided relevant supporting clinical features from the case.
- The student correctly recognized the importance of recent orthopedic surgery and reduced mobility as risk factors.
- The student appropriately mentioned assessing pretest probability and confirming the diagnosis with imaging.

Areas to improve

- Ensure the final diagnostic selection is explicitly filled out in the designated primary diagnosis field.
- Include structured differential considerations within the selected options rather than solely in the free text.

Safety notes

- Leaving the primary diagnosis field empty results in a zero score for that section despite correct free-text reasoning.
- Failure to check the corresponding hypothesis box impacts the score for selected propositions.

Full grading data

4. Management

Spontaneous reasoning

Rédaction

I would keep the patient under close clinical monitoring and regularly reassess her respiratory and hemodynamic status.

I would establish intravenous access and arrange a CT pulmonary angiogram if appropriate.

I would administer oxygen if she develops hypoxemia.

Because pulmonary embolism is strongly suspected, I would discuss early anticoagulation while carefully evaluating her recent surgery, bleeding risk and contraindications.

I would seek senior medical advice and monitor for any signs of clinical deterioration.

Selected answers

- reassess
- oxygen
- etiology
- followup
- reperfusion
- thrombolysis_all

Justification

—

Section confirmed : Yes

Generative AI feedback : 6 / 20

spontane: 5 · choix: 1 · priorites: 0 · justification: 0

Strengths

- Spontaneous text correctly identifies the need for close monitoring, intravenous access, and diagnostic imaging such as CT pulmonary angiogram.
- Spontaneous text appropriately mentions discussing early anticoagulation while evaluating the recent surgery and bleeding risks.

Areas to improve

- Avoid selecting dangerous or inappropriate management options from the checklist.
- Ensure critical interventions like timely anticoagulation and bleeding risk assessment are prioritized in the final choices.

Safety notes

- Severe penalty applied for selecting 'Give routine thrombolysis to every stable patient', which is medically dangerous and inappropriate for a hemodynamically stable patient with recent orthopedic surgery.

Full grading data

Final assessment

Generative AI : 55/100

- History taking : 18/20
- Initial assessment : 10/20
- Investigations : 14/25
- Diagnosis : 7/15
- Management : 6/20

This PDF is an archive of a simulated student attempt. Generative AI feedback is indicative and does not replace teacher assessment.